Hospital Bed Allocation: A Robust Optimization Approach

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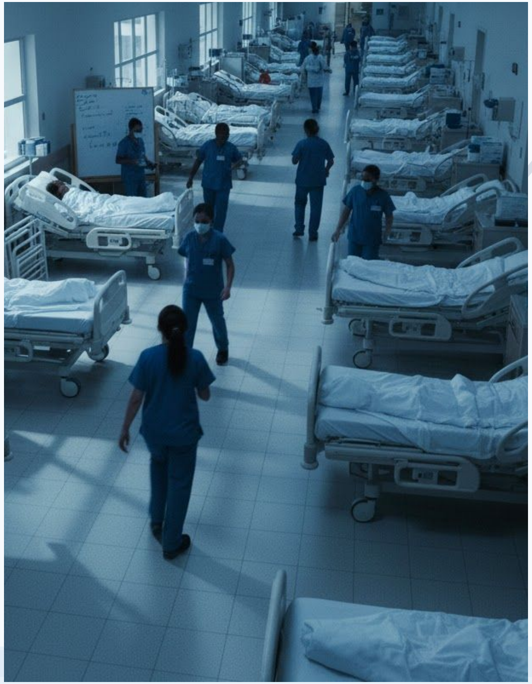
Project Motivation

- Motivated by the CDC's mission to improve national pandemic preparedness post COVID, we study the **hospital bed allocation problem** under uncertainty.
- Bed scheduling is a persistent complex operational challenge: capacity, clinical benefit, waiting time, and operational constraints all interact.
- Our project focuses on the **core drivers** of hospital bed allocation and formulates the right optimization objective.
- **Goal**: Decide *which patients to admit* and *when to admit* them under limited bed capacity and uncertain length of stay.



Project Scope

Our project aims to answer the following questions:

Optimal Admission Scheduleing

How should limited bed capacity be allocated over time to maximize clinical benefit?

Robust to Duration Uncertainty

How can the system be protected against unexpected overruns in treatment length?

Scenario Performance Normal vs Pandemic

How do the models behave under normal versus pandemic demand patterns?

Data

- Used **Hospital Beds Management** dataset (Kaggle)
- **1,000 synthetic patients** over a one-year horizon
- Includes key operational variables:
  - **Arrival time**
  - **Baseline length of stay**
  - **Service type**: ICU, Emergency, Surgery, General Medicine

- Based on the information from the dataset, we also generated the following **synthetic variables** to support our later formulation:
  - **Severity scores**: drawn from service-specific normal distributions.
  - **Admission benefits**: combining service-level base values with severity-based clinical benefit.
  - **Waiting penalties**: severity-weighted parameters to capture urgency-driven delay costs.

Data

- The variable notation is as follows:

Sets

- I: set of patients
- T: set of discrete time periods

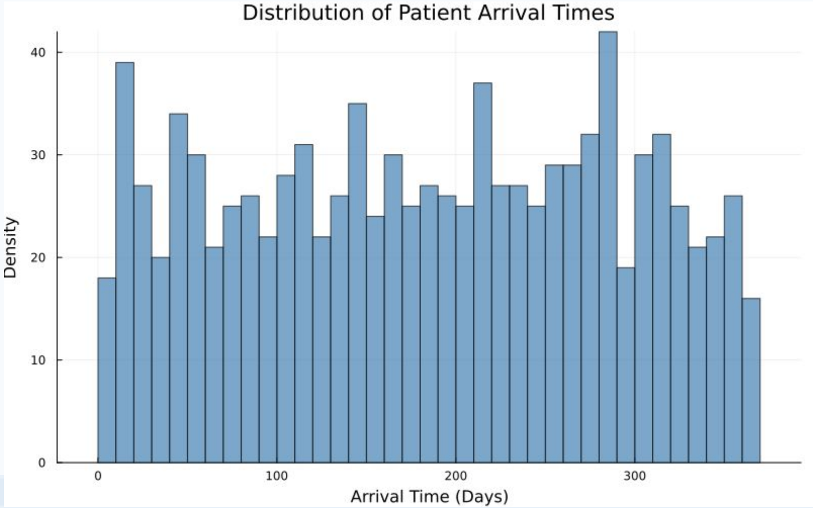
Parameters

- a_i: arrival time of patient i
- d_i: treatment duration
- s_i: severity
- w_i: admission benefit
- B: number of beds
- λ: weight of the waiting penalty

Decision Variables

- $x_{it} \in \{0, 1\}$: patient i starts treatment at time t
- $y_i \in \{0, 1\}$: patient i is admitted
- $q_{it} \in \{0, 1\}$: patient i waits at time t

- Analyzed weekly arrival frequency (7-day bins, matching average length of stay) to inform required bed capacity.



Method: Nominal Case

- The full formulation is shown on the right.
- Bed capacity fixed at **B = 15** to realistically mimic congestion with 1,000 annual patients.
- Objective **maximizes financial benefit while penalizing waiting time** proportional to patient severity.
- Model can **decline admissions** to prevent excessive waiting and untreated patients.
- Setup highlights **key trade-offs** between admissions, waiting time, and limited resources.

Objective

$$\max \sum_{i\in I} \sum_{t\geq a_i} w_i x_{it} - \lambda \sum_{i\in I} \sum_{t\in T} s_i q_{it}$$

Constraints

$$x_{it} = 0 \quad \forall i,\, t < a_i \qquad \text{(cannot start before arrival)}$$

$$\sum_{t\geq a_i} x_{it} = y_i \quad \forall i \qquad \text{(if admitted, must start exactly once)}$$

$$\sum_{i\in I} \sum_{\tau:\, t\in[\tau,\tau+d_i]} x_{i\tau} \leq B \quad \forall t \in T \qquad \text{(total occupied beds cannot exceed capacity)}$$

$$q_{it} = 0 \quad \forall i,\, t < a_i \qquad \text{(cannot wait before arrival)}$$

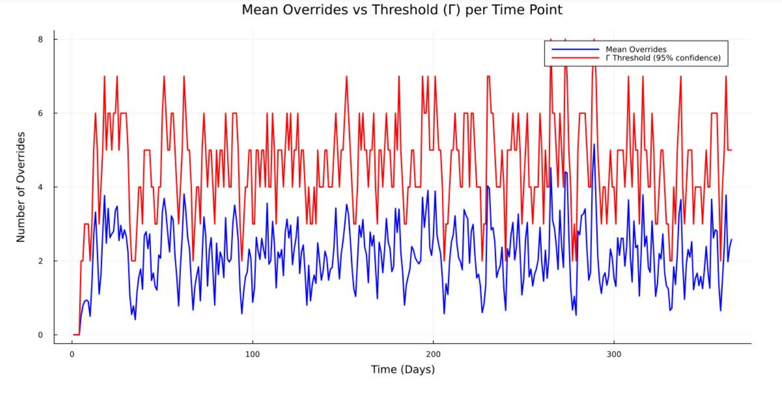
$$q_{it} \leq y_i \quad \forall i, t \geq a_i \qquad \text{(cannot wait if never admitted)}$$

$$q_{it} \leq 1 - \sum_{\tau=a_i}^{t} x_{i\tau} \quad \forall i, t \geq a_i \qquad \text{(cannot wait after admission)}$$

$$q_{it} \geq y_i - \sum_{\tau=a_i}^{t} x_{i\tau} \quad \forall i, t \geq a_i \qquad \text{(must wait before admission if admitted)}$$

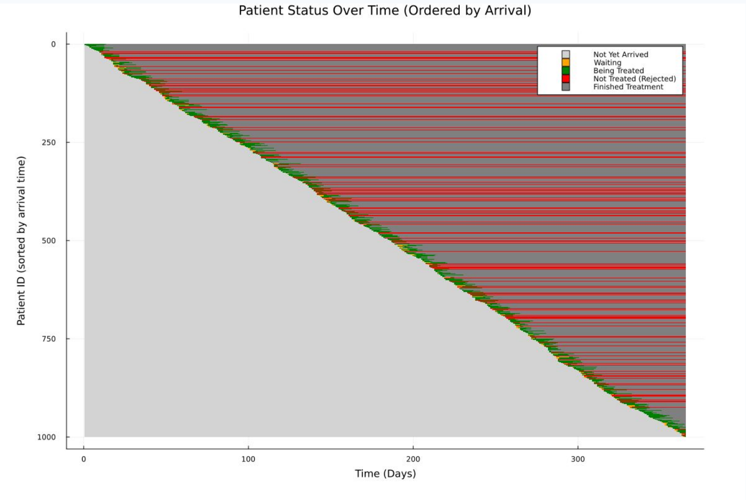
Method: Robust on Duration

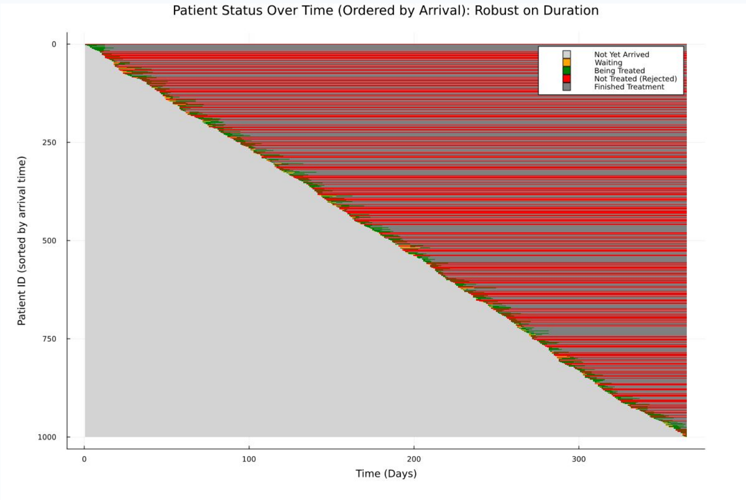
- Addressed **uncertain length of stay** by adding robustness to the model.
- Introduced a **time-varying buffer Γ□** representing reserved beds for potential overruns.
- Simulated **100 duration-overrun scenarios** (up to +5 days) to estimate risk.
- Set **Γ□ to the 95th percentile** of simultaneous overruns at each time period.



Result: Bed Assigment Schedule

Nominal vs Robust





Nominal model:

- High throughput; admits most patients
- Minimal waiting times
- Tight capacity usage with little buffer

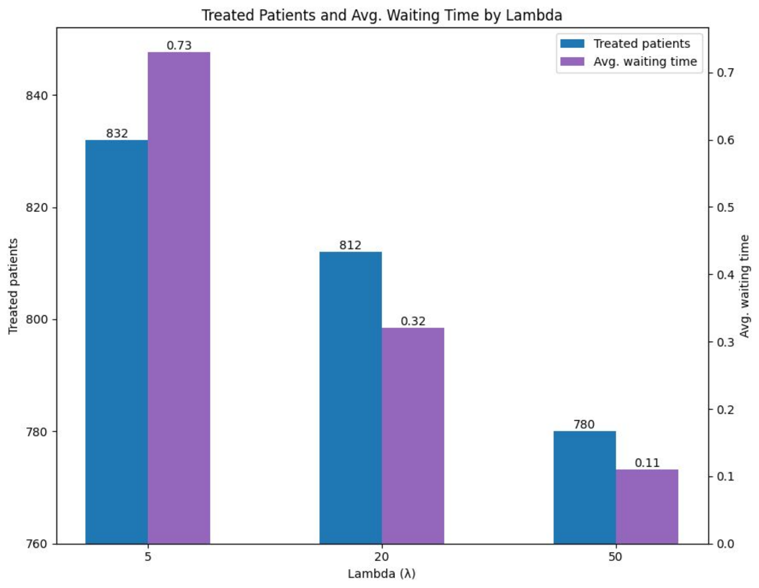
Robust model:

- Fewer admissions to preserve safety margin
- Prioritizes higher-severity patients
- Maintains buffer Γ□ to prevent overflows

Result Analysis

Sensitivity to Penalty Weight (λ)

- We also tried training the model with **different levels of penalty (λ)** for keeping patients waiting.
- Higher waiting penalties lead to fewer admissions and thus lower objective/benefit, while noticeably reduce waiting times.



Result Analysis

Stress Test Against Adversary

Stress Test of Robust Model (Overflow Protection):

Case	Duration Noise	Overflow Probability	Avg. Max Overflow
1	0–1 extra days	0.00	0.00
2	0–2 extra days	0.63	0.68
3	0–3 extra days	1.00	2.16

Table 5: Estimated overflow risk under different duration-uncertainty scenarios.

Strong performance stems from the small mean of the noise assumption, which the model is well-designed to protect against.

Result Analysis

Pandemic Scenario

- Pandemic dataset has **more high-severity arrivals**, shifting the severity distribution upward.
- Higher severity → **higher admission benefit**
- But higher severity also → **higher waiting penalty**

Metric	Pandemic	Normal
Treated patients	811	812
Rejected patients	189	188
Admission benefit	9.77×10^6	9.52×10^6
Waiting-penalty term	2.27×10^5	2.17×10^5
Objective value	9.54×10^6	9.30×10^6

Table 6: Comparison of model performance under pandemic vs. normal conditions ($\lambda = 20$).

- Increased benefit and increased penalty **offset** each other in the objective.
- As a result, **number of admissions remain similar**, whereas **objective value increases** in cases of pandemic

Key Findings and Impact

Concept proven:

- **Severity-based prioritization** works as intended.
- Deterministic model performs well **under stable conditions**.
- Robust model **protects capacity** but **reduces throughput**.

Business Impact:

- Provides a **reliable bed-scheduling tool** that balances financial value and patient wellbeing
- Robust formulation significantly **reduces overflow risk**, ensuring admitted patients receive timely treatment
- Demonstrates the operational trade-off: hospitals must balance **overflow vs. reliability**

Future Steps

- Incorporate **real hospital data** to replace synthetic inputs and improve realism.
- Extend **robustness to other uncertain variables**, such as arrival times and patient volumes.
- Develop a **dynamic program** with multiple stage decision making to replace the fixed year-long scheduling approach.
- With more computing power, **scale up patient volume** in the pandemic scenario and test more skewed severity distributions.
- Simplify or reformulate the model to potentially reduce high computational cost (30+ minutes per run).


Thank You!